

Do LLMs Reason Consistently Across Programming Languages? An Evaluation Through Formal Specification Inference

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Abstract

Large Language Models (LLMs) have demonstrated strong capacity in code generation. However, their ability in reasoning the semantic meaning of program still remains unclear. Formal specification inference provides a rigorous means of investigating their capacity by requiring models to reason program behavior using precise logical constraints. This project investigates whether LLMs exhibit consistent semantic reasoning when equivalent program behavior is expressed in different programming languages. We construct paired implementations of semantically equivalent programs from Java to C and evaluate LLM generated formal specifications expressed in Java Modeling Language (JML) and ANSI/ISO C Specification Language (ACSL). Generated specifications are evaluated using formal verification, with particular attention to their consistency with program implementations and their ability to characterize program behavior. We compare verification outcomes across paired implementations and analyze cases where semantically equivalent programs produce different specification-generation outcomes. The study aims to determine the extent to which LLM formal reasoning reflects language-independent program semantics rather than language-specific representations, and to identify factors contributing to cross-language differences in specification generation.